

Course Code: CS3001	Course Name: Computer Networks
Instructor Name: Dr. Farrukh Salim , Dr. Aqsa Aslam, Mr. Shoaib Raza and Ms. Eman Shahid	
Student Roll No:	Section:

Instructions:

- Return the question paper. Do not write anything on question paper, except your Roll # & Section #.
- Read each question completely before answering it. There are **5 questions and 2 pages**.
- In case of any ambiguity, you may make assumptions. However, your assumptions should not contradict any statement in the question paper.
- All the answers must be solved according to the sequence given in the question paper.
- This paper is subjective. Write the answers only on answer sheet.

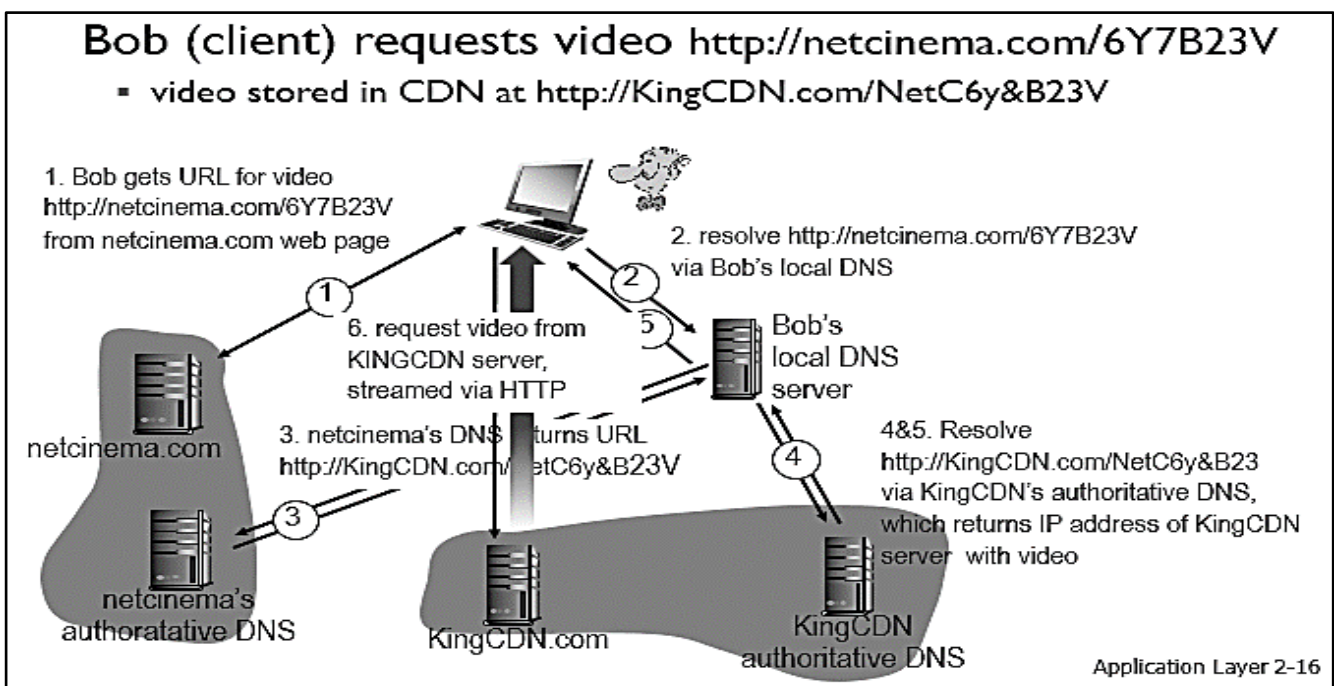
Time: 60 minutes.**Max Points:** 50**Question 1:**

[10 points]

Suppose Akamai is a company that operates Content Distribution Networks and sells its services to other companies. Assume that CNN.com has a contract with Akamai, in which Akamai will distribute all of the JPEG files on the CNN.com's web pages. All non-JPEG files will continue to be distributed directly by the CNN.com. Explain with the help of a diagram how CNN.com and Akamai use DNS to redirect your requests for JPEG files on CNN's site so that the files are retrieved from an Akamai server 'near' to you.

Answer:

This is similar to the scenario we studied in the class. Here Netcinema.com is CNN.com and KingCDN.com is Akamai. See the diagram below



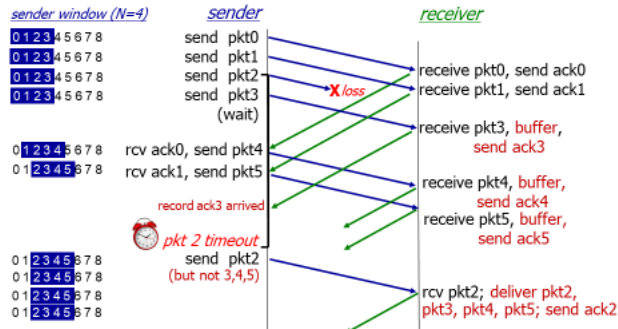
Question 2:

[10 points]

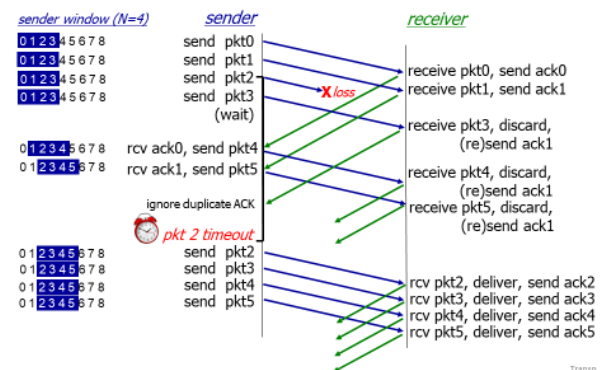
Let's suppose a window size of 4 and sequence number range from 0 to 8. Apply both Go-Back-N (GBN) and Selective Repeat (SR) for sender and receiver events and actions. Assume that 2nd packet in the window is lost. Show the SR and GBN operations using the diagram in the presence of the lost packet.

Solution:

Selective Repeat in action



Go-Back-N in action



Question 3:

points]

[4 * 2.5 = 10

Considering the following scenario shown in Figure 1 and briefly answer the questions below.

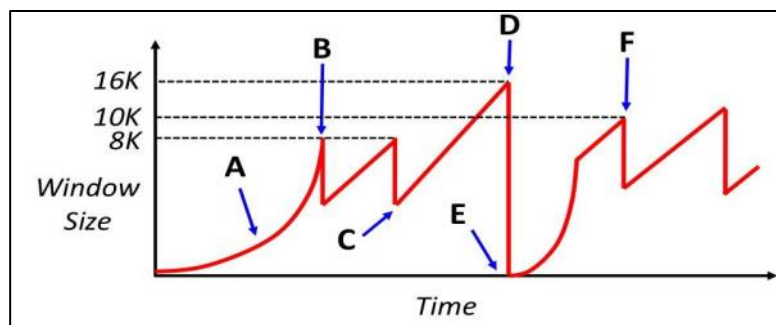


Figure 1. TCP Reno's congestion window

- Explain the event at B which occurs that causes the sender to decrease its window.
Triple Duplicate Ack
- Does the event at B necessitate that the network discarded a packet? Explain.
No, It could be due to reordering. Due to queuing and asymmetric paths.
- Explain the event at D which occurs that causes the sender to decrease its window.
Timeout
- Does the event at D necessitate that the network discarded a packet? Explain
No. Congestion in either direction could cause $RTT > RTO$ (retransmission timeout)

Question 4:

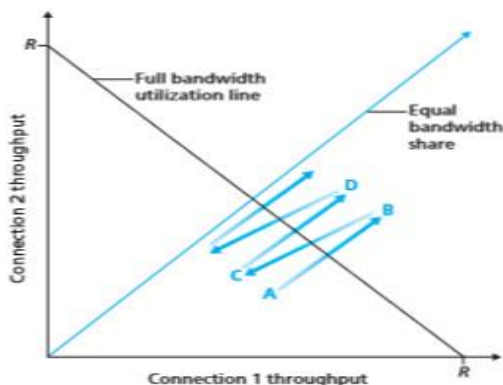
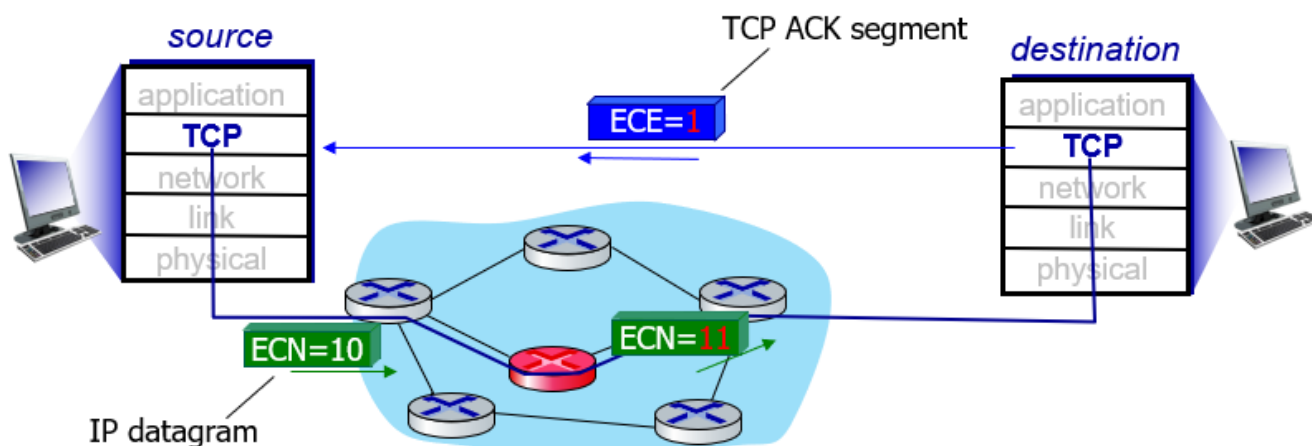
[5 + 5 = 10 points]

Explain with the help of a diagram:

- (a) How the fairness is ensured if two TCP connections share the bandwidth between each other?
(b) how explicit congestion notification works. How it notifies the sender and receiver about network congestion?

Answer (a):

By Using AIMD Control law.

**Answer (b):****Question 5:**

[10 points]

Using the TCP Receiver, ACK Generation Recommendation discussed in class, draw the message sequence (timing) diagram, showing the segments and their respective acknowledgments for the following, as in Figure 2:

Received Timing (sec):	0.5	0.75	1	2	3	5.25	5	3.75	4
Byte Number:	0	1	2	3	4	5	6	7	8
Packets:	☐	☐	☐	☐	☐	☐	☐	☐	☐

Figure 2. Arrival time and sequence of packets at the TCP receiver

Assuming:

- Timeout interval set as 4 seconds.
- Both the sender (Host A) and receiver (Host B) are connected, with a propagation delay of 0.5s.
- All other delays (like transmission, queuing and nodal processing delays) to be negligible and thus considered as zero.
- Each packet is of one byte only and the sequence number start with zero.

